MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS)

THIRDSEMESTEREXAMINATION

(2024ADMISIONONWARDS)

MG3DSCMOS200 IoTfor Electronics

Duration:1½hours **MaximumMarks:50**

.

**Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Skill(S),Interest (I)andAppreciation(Ap)*

PartA

MultipleChoiceQuestions

Answer **All** Questions

Eachquestioncarries1mark

1	WhichofthefollowingisnotacharacteristicofIoTdevices? (a)Interconnectivity (b)Scalability(c)Real-timemonitoring(d)Manualdataentry	K	1
2	WhichofthefollowingisanexampleofanIoTprotocolforwirelesssensornetworks? (a)MQTT(b)ZigBee(c) FTP (d) HTTP	K	2
3	WhichfunctioninArduinoisusedtooutputdatatotheSerialMonitor? (a)Serial.read()(b)Serial.write()(c)Serial.print()(d)Serial.begin()	K	3
4	OICarchitectureismainlyfocusedon: (a)Databasemanagement (b)Deviceinteroperabilityand standardization (c)Videostreamingoptimization(d)Socialmediaapplications	U	2
5	WhatdoesthebasicIoTblockdiagramtypicallyconsistof? (a)Sensing,Communication,Processing,Application (b)Storage,Retrieval,Encryption,Execution (c)Sensing,Storing,Printing, Sharing (d)Input,Output,Feedback,Loop	K	1
6	IoTivityisusedto: (a)CreatewebsitesforIoTdevices (b)EnablecommunicationandresourcediscoverybetweenIoTdevices (c)Convertsensordatainto images (d)MonitorWi-Fitraffic	U	2
7	DHT11isusedtomeasure: (a)Lightintensity(b)Temperatureandhumidity(c)Airpressure(d)Soilmoisture	K	3
8	WhichofthefollowingisakeyfunctionoftheIoTivityresourcemodel? (a)Printing IoTdevicedata (b)RepresentingandmanagingIoT deviceresources (c)Encryptingsocialmediamessages (d)Sendingemailsautomatically	U	2
9	WhichofthefollowingisachallengeinIoT? (a)Increasedautomation (b)Privacyandsecurityissues (c)Energyefficiency (d)Reducedcostofdevices	K	1
10	WhichtypeofsignalisrequiredtocontrolaservomotorfromESP32? (a)ConstantDCvoltage(b)PWM(c)Analogsignalonly(d)ACsignal	U	3

1 * 10 = 10

PartB

Short Answer Questions

Answer any **4** Questions

Eachquestioncarries5marks

11	ExplainthekeycomponentsofIoTwithsuitableexamples.	U	1
12	ComparetheWebofThings(WoT)withtheInternetofThings(IoT).	U	2
13	WhatarethreemaincharacteristicsofIoTdevices?	K	1
14	HowcanarelayandDCgearmotorbecontrolledusingESP32GPIOpins?	U	3
15	DescribetheworkingprincipleofaPIRsensorandgiveoneapplication.	K	3
16	GivetwoexamplesofIoTapplicationsinindustryandexplaintheirbenefits.	U	2

4 * 5 = 30

PartC

Essay Questions

Answerany**2**Questions

Eachquestioncarries10marks

17	Discussthemajortrends,technologydrivers,andbusinessdriversofIoT. Explain how they impact IoT adoption in modern industries.	U	1
18	DiscusstheroleofIoTprotocolsandopen-sourceIoTarchitecturesinachieving interoperability and scalability in IoT ecosystems.	U	2
19	Explain with suitable examples how serial communication, analog and digital I/O are used in controlling devices.	K	3
20	Explainthe importance ofprotocol standardization in IoT and also discuss the roles of M2M and WSN protocols.	U	2

2 * 10 = 20